

Skin Lesion Segmentation in Dermoscopic Images

A comparison of three classical computer-vision methods

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Abstract

Melanoma is the most aggressive form of skin cancer, and its prognosis depends closely on how early it is detected. Automated delineation of the lesion is the first step of any computer-aided diagnosis system, since every downstream clinical measurement (asymmetry, border irregularity, colour) is computed from that border. This report presents a segmentation pipeline built *without* deep learning, under the explicit constraint of using classical image-processing techniques only. After a preprocessing stage removing the dermoscope frame and body hair, three segmentation methods are implemented from their source publications and compared on 20 annotated dermoscopic images: multi-channel Otsu thresholding, LBP Clustering, and Statistical Region Merging. LBP Clustering gives the best raw masks, and a shape-aware convex hull raises every method further. We report where each method fails and why.

Introduction

Melanoma is a major public health concern: in the United States alone, roughly 40,000 to 50,000 diagnoses per year lead to about 10,000 deaths. Survival depends strongly on how quickly the lesion is identified and treated, so any technological improvement in screening tools can reduce mortality.

Dermoscopy is the reference imaging technique. By eliminating surface reflections it reveals subsurface structures, improving diagnostic sensitivity over a simple clinical examination. Interpreting these images nevertheless remains difficult and subject to inter-operator variability, even among experienced dermatologists. Computer-aided diagnosis (CAD) systems therefore matter: they offer a more objective, standardised and reproducible assessment. Such systems rest on segmentation algorithms whose role is to delineate the pathological area precisely.

This project automates that critical isolation step for pigmented lesions in dermoscopic images, under one specific constraint: classical image-processing techniques only, with no deep learning. The constraint is deliberate. A segmentation network needs annotated training data that a twenty-image study does not have, and its failures are hard to attribute; a classical pipeline runs without training and each of its stages can be inspected in isolation, which is what allows the failure analysis in Section 3.

The report first details the preprocessing needed to clean the images of their artefacts, then presents the implementation and comparison of three segmentation methods, and finally evaluates them against expert ground truth.

1 Preprocessing of dermoscopic images

1.1 Frame removal

Many dermoscopic images show a dark frame around the useful area, caused by the acquisition device or by a physical mask (vignetting). These black borders can be mistaken for the lesion, or disturb the segmentation algorithms by creating artificial contrast. Removing the frame before segmentation is therefore essential.

1.1.1 First approach: rectangular scanning

We initially implemented an iterative cropping algorithm scanning rows and columns from the image borders inwards. The algorithm computed the proportion of black pixels on each line; as soon as that proportion fell below a threshold (indicating skin), scanning stopped and defined the crop limits.

Observed limitations. This method implicitly assumed the region of interest was rectangular and aligned with the image borders. On several images of our dataset, however, dermoscopes produce a circular field of view. Dark corners survived the crop, which led us to a more robust approach accounting for the circular geometry.

1.1.2 Second approach: morphological segmentation and circular masking

Instead of removing borders line by line, the geometric approach identifies the central skin area as an *object* and masks out everything else. The algorithm assumes the illuminated skin forms a bright disc near the image centre, and proceeds as follows:

1. **Border analysis.** A preliminary inspection of a band covering 5 % of the image width gives the proportion of black pixels. If that proportion is low the image is considered frameless and left untouched. That matters, since cropping a frameless image would only destroy data.
2. **Global thresholding and connected components.** The image is converted to grayscale and thresholded to separate the bright skin from the dark background. Connected-component analysis retains the largest region, discarding noise and isolated reflections.
3. **Geometric modelling.** The main component is approximated by a disc whose centroid and radius are computed. The radius is shrunk by 10 % to stay clear of the blurred rim.
4. **Masking and cropping.** Everything outside the disc is replaced by white (255) rather than black, so the discarded area cannot later be mistaken for a dark lesion. A rectangular crop around the disc follows.

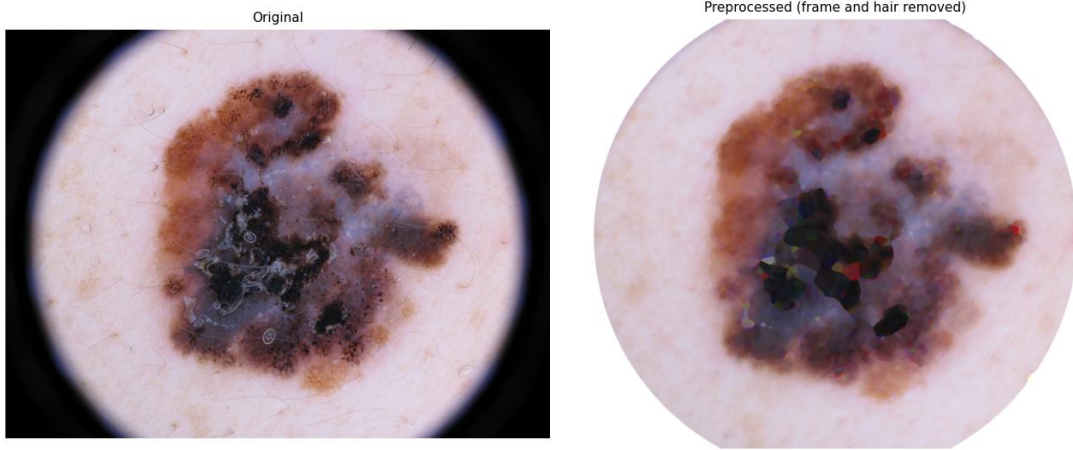


Figure 1: Preprocessing. Left: the raw dermoscopic image with its dark frame. Right: after frame removal and hair removal. The corners are whitened and the image cropped to the dermoscope disc.

Discussion. This step proved robust on the majority of standard images. Simple thresholding can nevertheless show its limits when the frame is partially reflective or the illumination strongly heterogeneous. Integrating shading-correction techniques [2], which model illumination variation by a quadratic function, would be a natural improvement. For this project the current method was sufficient.

One consequence of whitening the corners deserves emphasis, because it caused a real failure later in the pipeline: the whitened area is *synthetic*, not skin. Any method fitting a global colour model over the whole frame will let those pixels form a cluster of their own. Section 2.2 returns to this.

1.2 Hair removal

Dermoscopic images often contain hairs crossing the lesion or the surrounding skin. These artefacts mislead segmentation algorithms, which read them as contours or as internal lesion structures.

Our approach is based on the reference algorithm DullRazor [1], which detects thin dark structures by morphological closing.

1.2.1 First approach: naive implementation and its limits

We first tested a standard implementation using thin linear structuring elements (one pixel wide) and a high global threshold (250) on the difference image.

Analysis of the failure. This showed critical limits on images with thick or low-contrast hairs.

1. **Scale problem.** The structuring element being thinner than some hairs, the closing operated *inside* the hair without removing it, making detection impossible.
2. **Threshold problem.** A threshold of 250 demanded near-maximal contrast (black on white). The actual intensity difference between a brown hair and skin is often between 20 and 50, so the resulting binary mask was mostly empty and no interpolation could take place.

1.2.2 Second approach: a robust pipeline

The improved version detects complex structures without degrading skin texture:

1. **Multi-directional closing.** Rather than thin lines, we use rectangular structuring elements of 50×10 and 10×50 pixels, taking the pixelwise maximum of the two closings so that hairs at any orientation are caught.
2. **Detection and dilation.** The hair mask is the absolute difference between the original and the closed image, thresholded at 40 instead of 250. Although lower, this threshold still ignores weak skin-texture variation while capturing hairs, thanks to the strong response produced by the wide structuring elements. The mask is then dilated by a 20×20 square to include the blurred edges of thick hairs.
3. **Interpolation and smoothing.** Masked areas are reconstructed by nearest-neighbour grid interpolation from the immediate neighbourhood, and a final median filter with a 20×20 window blends the corrections into the surrounding healthy skin.

A coverage test guards the whole stage: if the eroded hair response covers less than 20 % of the valid area, the image is returned untouched. Inpainting a hairless image only blurs the very border the segmentation is looking for.

2 Segmentation methods

Having cleaned the artefacts, the critical step is isolating the pigmented lesion from healthy skin. We implemented and compared three representative families of algorithms: a histogram-based method (Otsu), a texture-based method (LBP Clustering) and a region-merging method (SRM).

All three compute on a 256×256 copy of the image and upscale the resulting mask. The lesion border is a large-scale structure, so this loses no useful detail, and it keeps the three methods comparable by giving them the same amount of information.

2.1 Method 1: multi-channel Otsu thresholding

2.1.1 Theoretical principle

Otsu’s method is the classical approach to image binarisation. It assumes the intensity histogram is bimodal (background and object) and searches for the threshold T minimising intra-class variance, equivalently maximising the inter-class variance

$$\sigma_B^2(T) = \omega_0(T) \omega_1(T) [\mu_0(T) - \mu_1(T)]^2,$$

where ω_0, ω_1 are the class probabilities (skin/lesion) and μ_0, μ_1 their respective means.

2.1.2 Multi-channel combination

The performance of a global threshold depends drastically on the colour channel used [5]. A lesion may be invisible in the red channel, where skin and melanin reflect light similarly, yet strongly contrasted in blue.

Otsu’s threshold is therefore computed independently on the red, green and blue channels, and the three binary masks combined as

$$M = (M_R \wedge M_G) \vee M_B.$$

A pixel is lesion when red *and* green agree, or when blue alone is confident. The asymmetry is deliberate: melanin absorbs short wavelengths preferentially, so blue carries the strongest

melanin contrast and is trusted on its own, whereas red and green are required to corroborate each other.

The resulting mask has ragged borders. It is therefore used as the initial level set of a morphological Chan-Vese active contour, run for five iterations, which relaxes the border onto the actual intensity boundary.

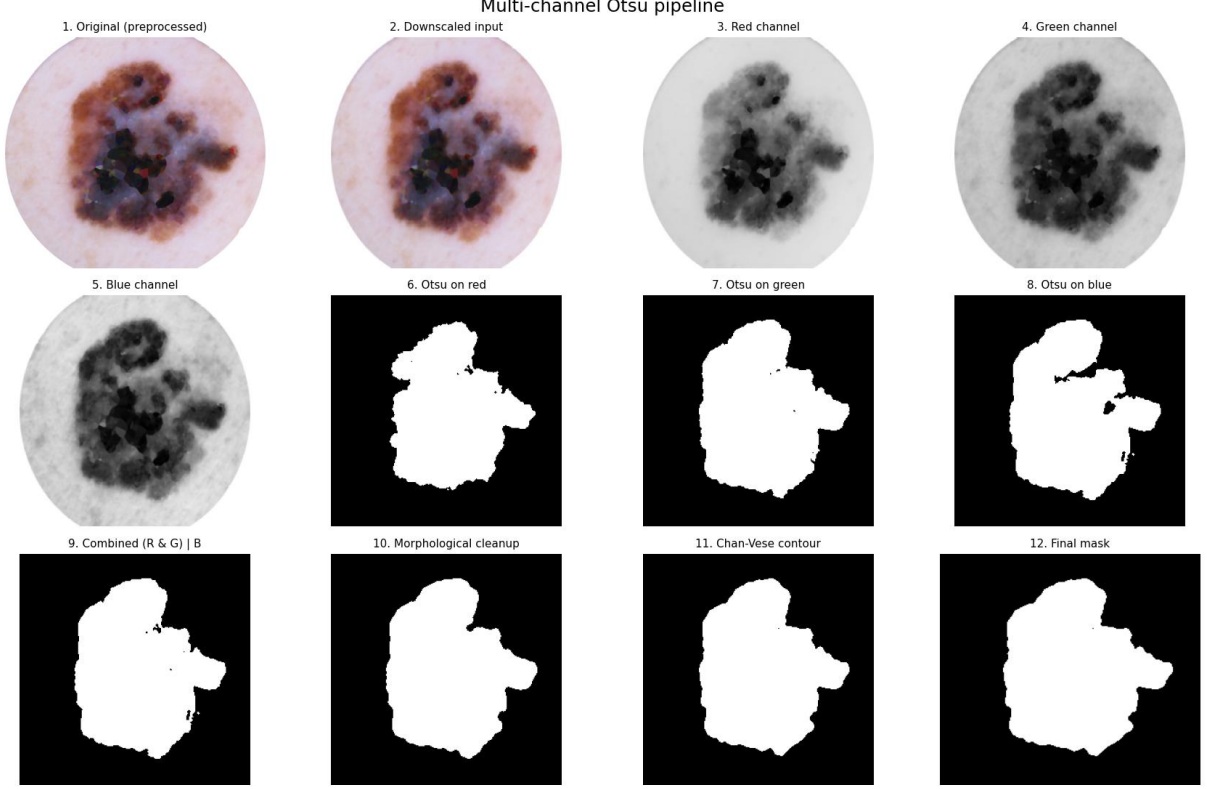


Figure 2: The multi-channel Otsu pipeline, stage by stage: the three colour channels, their individual Otsu masks, the combination, the morphological cleanup and the Chan-Vese refinement.

2.1.3 Observed limitations

Despite its speed, the method shows intrinsic limits on our dataset. Being global, it fails when the image presents strong vignetting or cast shadows, the shadow being segmented as part of the lesion. For early or very pale melanomas the histogram is not clearly bimodal, and the computed threshold merges the lesion with the surrounding skin, causing critical under-segmentation.

2.2 Method 2: LBP Clustering

Simple thresholding fails when the lesion colour is too close to that of the skin. We therefore implemented a texture-based method following Pereira et al. [3], whose founding assumption is that healthy skin has a smooth, regular texture whereas a lesion has a chaotic one, with its pigment network, globules and dots.

2.2.1 Local Binary Patterns and pattern-subset binarisation

The luminance Y is computed according to ITU-R BT.601,

$$Y = 0.299 R + 0.587 G + 0.114 B.$$

For each pixel, the Local Binary Pattern with $P = 8$ neighbours at radius $R = 1$ encodes the sign of the difference between each neighbour and the centre, packing the eight bits into a code in $[0, 255]$. The comparison is strict, matching the paper’s $s(u) = \mathbf{1}[u > 0]$.

The key step is the binarisation on a *subset* of patterns. Code 0 (no neighbour brighter) and the eight powers of two (exactly one brighter neighbour) describe locally smooth neighbourhoods; every other code describes texture. Mapping the smooth subset to 0 and all others to 1 yields a texture map whose ones concentrate inside the lesion.

2.2.2 Pseudo-RGB construction and the pinkness criterion

The binary texture map is smoothed by a Gaussian filter ($\sigma = 3$) and normalised into a continuous field $L \in [0, 1]$. The method then builds a *synthetic* image by stacking

$$\text{pseudo-RGB} = [L, Y, L],$$

and converts it to CIE $L^*a^*b^*$, keeping only the chromaticity components (a^*, b^*) . This is the elegant part of the method: a texture quantity and a luminance quantity are deliberately placed in a colour space so that a *colorimetric* criterion can separate them.

A k -means++ clustering with $k = 2$ partitions the pixels in that (a^*, b^*) plane. Which of the two clusters is the lesion is decided without supervision by the *pinkness* score

$$\mathcal{P} = \text{mean}[\max(a^*, 0) - \min(b^*, 0)],$$

the cluster with the higher score being retained. A morphological opening finishes the mask.

2.2.3 Interaction with frame removal

Applying the method as published to our cropped images failed completely on two of them, with a Dice score of exactly zero. The cause is the interaction announced in Section 1.1: frame removal whitens the corners, and a white pixel has $L = 0$ texture with $Y = 255$, so in the pseudo-RGB it becomes *pure green*, an extreme point in the (a^*, b^*) plane. Those synthetic pixels captured one of the two clusters outright, and k -means separated disc from corners instead of lesion from skin.

Restricting the clustering to the valid disc area resolves it, and the two failing images rose from 0.000 to 0.915 and 0.957. The lesson generalises beyond this project: a preprocessing step that fabricates pixel values must publish which pixels it fabricated.

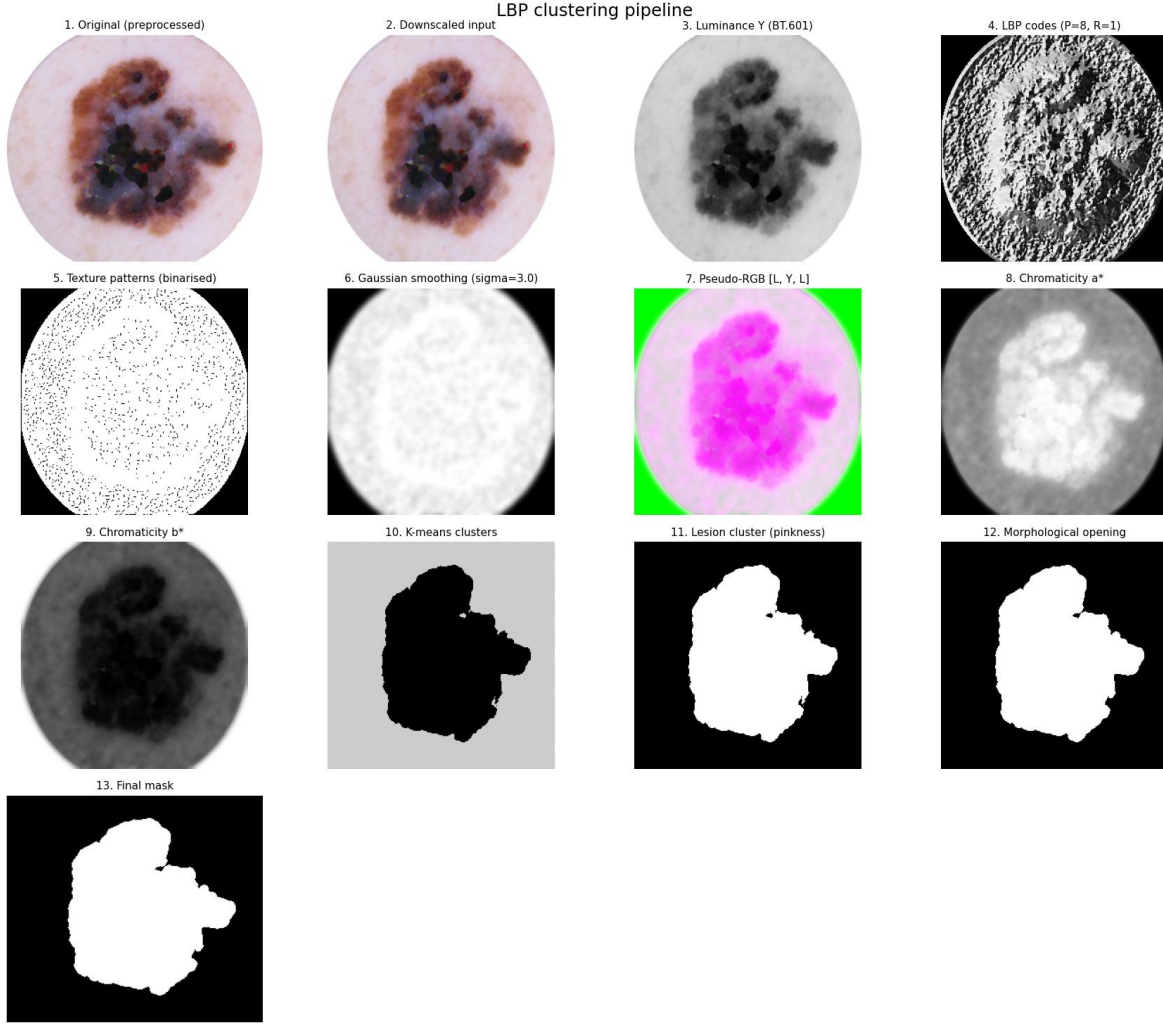


Figure 3: The LBP Clustering pipeline: luminance, LBP codes, the binarised texture patterns, the smoothed field L , the pseudo-RGB $[L, Y, L]$, its chromaticity components, the k -means clusters and the cluster selected by the pinkness criterion.

2.3 Method 3: Statistical Region Merging

SRM [4] is a region-growing technique. Every pixel starts as its own region, and adjacent regions are merged iteratively when they are statistically similar.

2.3.1 The merging predicate

Pairs of 4-connected neighbouring pixels are sorted by increasing absolute intensity difference and visited in that order, so the most obviously similar neighbours merge first. Two regions R_1 and R_2 are merged when

$$(\overline{R_1} - \overline{R_2})^2 < b(R_1) + b(R_2), \quad b(R) = \frac{g^2}{2Q|R|} \left(\ln(1 + |R|) + 2 \ln(6n) \right),$$

where $\overline{R}$ is the region mean, $|R|$ its size, n the pixel count and $g = 1$ for intensities in $[0, 1]$. The bound tightens as regions grow: early merges are cheap, late merges must be well supported. The scale parameter Q controls permissiveness, a smaller Q yielding fewer, larger regions. Regions are maintained in a union-find structure with path compression and union by rank.

The implementation follows the algorithm directly rather than calling a library, and the smoothing is applied to the grayscale image, not to the colour image: a scalar Gaussian σ on

an $(H, W, 3)$ array also blurs *across* the channel axis, mixing red into blue. We measured this mistake to cost 0.35 Dice on a single image (ISIC_0000146: 0.440 before correction, 0.792 after) before it was found.

2.3.2 From regions to a lesion

SRM produces an over-segmentation, not a lesion. Each region is scored on three criteria, combined as $0.5c + 0.3z + 0.2s$:

- **Contrast** c against skin. The skin reference is the 70th percentile of a band along the image border, following the observation of Zortea et al. [6] that this band is almost entirely healthy skin after cropping. The blue channel is used, for the melanin-absorption reason given in Section 2.1.
- **Centrality** z , a prior that the lesion lies near the centre.
- **Saturation** s , the HSV saturation above the skin median, which detects erythematous areas that are bright but pathological.

Polychrome lesions, a dark core with a brown periphery, are split by the over-segmentation, and keeping only the best region under-estimates the area. We therefore merge the winning region with every region scoring above 60 % of the maximum *and* genuinely darker or more saturated than skin. The second condition prevents the mask from creeping into shaded skin.

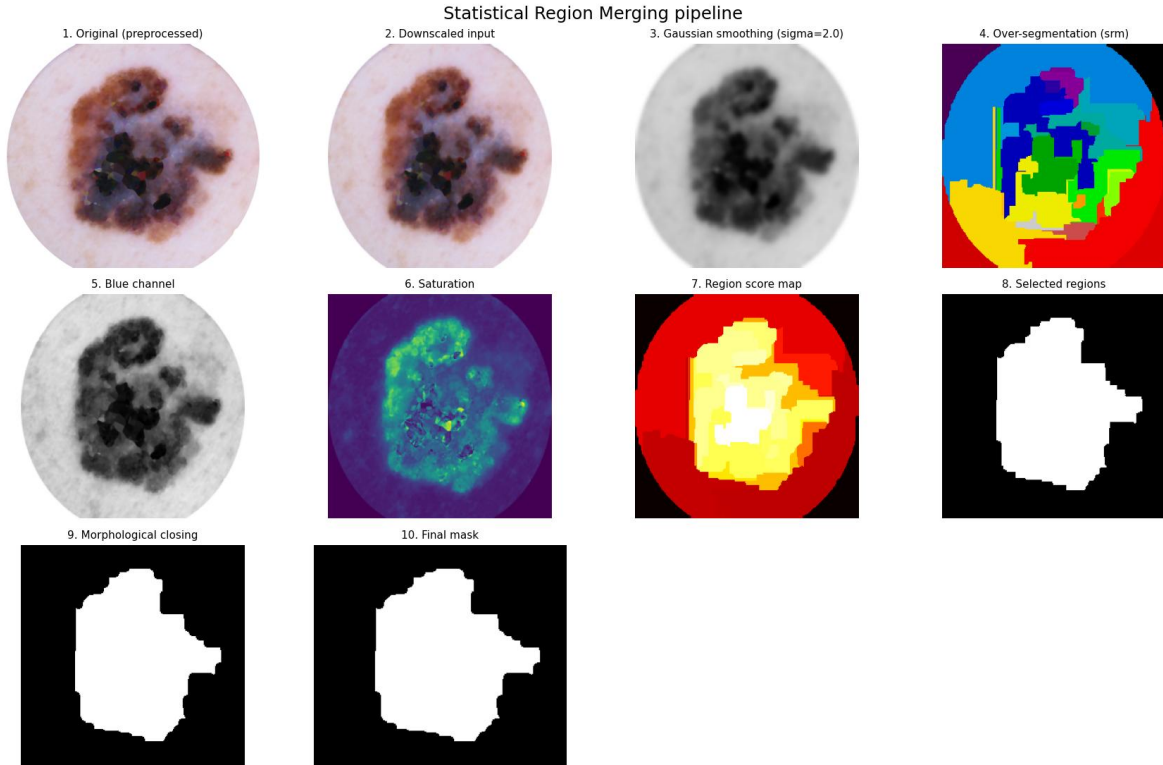


Figure 4: The SRM pipeline: smoothing, the union-find over-segmentation, the blue and saturation channels driving the scoring, the resulting score map, the selected regions and the final mask.

2.4 Post-processing: the smart convex hull

Using a convex hull to smooth borders is common practice but risky. On some images a small residual artefact in a corner produced an aberrant triangular mask spanning the whole image.

Before computing the hull, our `smart_hull` filter analyses the connected components of the mask. It always keeps the largest, and keeps a satellite component only if it is large relative to the main one (more than 3 % of its area) or close to it (within 2.5 equivalent radii). Distant specks of noise are discarded, so the convex shape stays meaningful.

The same post-processing (hole filling, an inversion guard for masks that selected skin rather than lesion, and this hull) is applied identically to all three methods, so that the comparison isolates the segmentation itself.

3 Results and discussion

3.1 Quantitative analysis

Each predicted mask is compared to the expert ground truth by the Dice coefficient. Table 1 gives the means over the 20 images.

Method	Dice (mask)	Dice (with convex hull)
Multi-channel Otsu	0.844	0.897
LBP Clustering	0.871	0.884
Statistical Region Merging	0.817	0.866

Table 1: Mean Dice coefficient per method.

Two trends stand out.

The convex hull helps, but not uniformly. It raises Otsu and SRM substantially, because both tend to fragment the lesion and the hull glues the pieces back together. It helps LBP far less, whose masks are already connected, and on individual nevi it actively hurts. Convexity is an *assumption* about lesion shape, and a lesion with a concave border violates it; reporting the hull as a systematic improvement, as we did in an earlier version of this work, was an over-generalisation from the mean.

Texture beats intensity on raw masks. LBP Clustering produces the best masks before post-processing. Where Otsu hesitates on a blurred boundary, the texture criterion still discriminates the lesion by its surface roughness even when its colour is close to skin.

3.2 Melanoma versus nevus

Method	Melanoma		Nevus	
	mask	+ hull	mask	+ hull
Multi-channel Otsu	0.823	0.877	0.866	0.917
LBP Clustering	0.850	0.896	0.893	0.872
Statistical Region Merging	0.782	0.845	0.852	0.887

Table 2: Mean Dice coefficient per method and per diagnosis.

All three methods score lower on melanomas than on nevi. This is the clinically expected direction rather than a defect: irregular, poorly defined and asymmetric borders are precisely what characterises melanoma, and precisely what the ABCD rule looks for. A segmentation method that performed equally well on both would be suspicious.

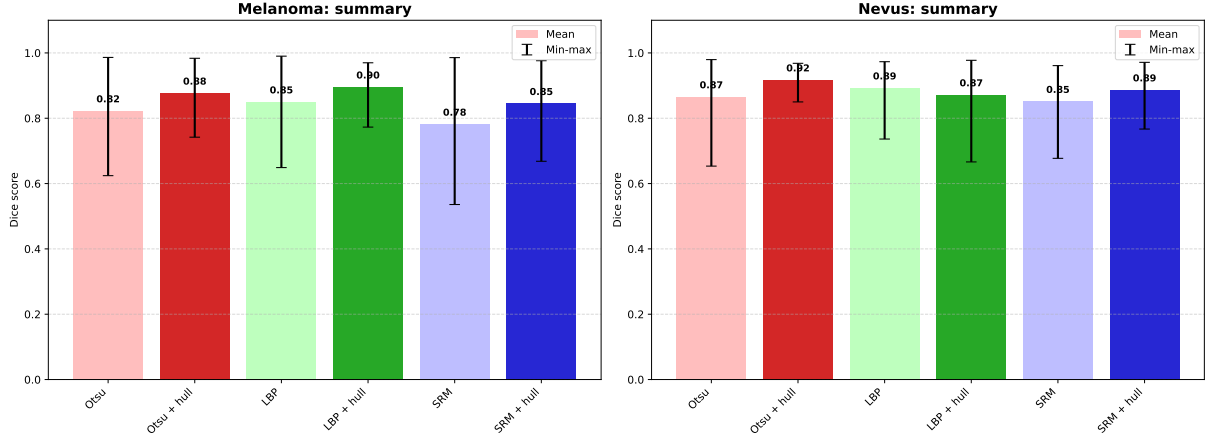


Figure 5: Mean Dice per method with min-max whiskers, melanoma (left) and nevus (right). The spread is markedly wider on melanomas.

3.3 Failure analysis

The per-image scores in Table 3 make the failure modes explicit.

Two images drag every method down: ISIC_0000030 and ISIC_0000049 are low-contrast lesions where neither an intensity nor a texture cue separates lesion from skin cleanly. No amount of parameter tuning rescued them, and we consider them the honest boundary of the classical approach. These are the cases where a learned model, given enough annotated data, would win.

Image	Class	Otsu	+hull	LBP	+hull	SRM	+hull
ISIC_0000030	melanoma	0.67	0.76	0.68	0.79	0.59	0.72
ISIC_0000046	melanoma	0.89	0.93	0.82	0.89	0.72	0.77
ISIC_0000049	melanoma	0.62	0.74	0.65	0.77	0.54	0.67
ISIC_0000140	melanoma	0.91	0.93	0.91	0.93	0.91	0.93
ISIC_0000142	melanoma	0.90	0.92	0.93	0.95	0.91	0.93
ISIC_0000143	melanoma	0.71	0.77	0.77	0.83	0.70	0.75
ISIC_0000145	melanoma	0.99	0.98	0.99	0.97	0.99	0.98
ISIC_0000146	melanoma	0.91	0.93	0.93	0.95	0.79	0.88
ISIC_0000150	melanoma	0.81	0.94	0.89	0.95	0.81	0.93
ISIC_0000151	melanoma	0.83	0.87	0.93	0.93	0.88	0.91
ISIC_0000000	nevus	0.90	0.92	0.94	0.94	0.88	0.91
ISIC_0000001	nevus	0.98	0.96	0.88	0.86	0.95	0.93
ISIC_0000008	nevus	0.93	0.97	0.97	0.98	0.95	0.97
ISIC_0000019	nevus	0.87	0.91	0.90	0.93	0.85	0.89
ISIC_0000024	nevus	0.65	0.90	0.74	0.83	0.68	0.77
ISIC_0000042	nevus	0.80	0.86	0.84	0.90	0.72	0.79
ISIC_0000045	nevus	0.95	0.96	0.96	0.95	0.89	0.90
ISIC_0000080	nevus	0.94	0.96	0.97	0.96	0.96	0.96
ISIC_0000095	nevus	0.86	0.90	0.91	0.67	0.86	0.90
ISIC_0000112	nevus	0.76	0.85	0.81	0.71	0.79	0.84

Table 3: Dice coefficient per image and per method.

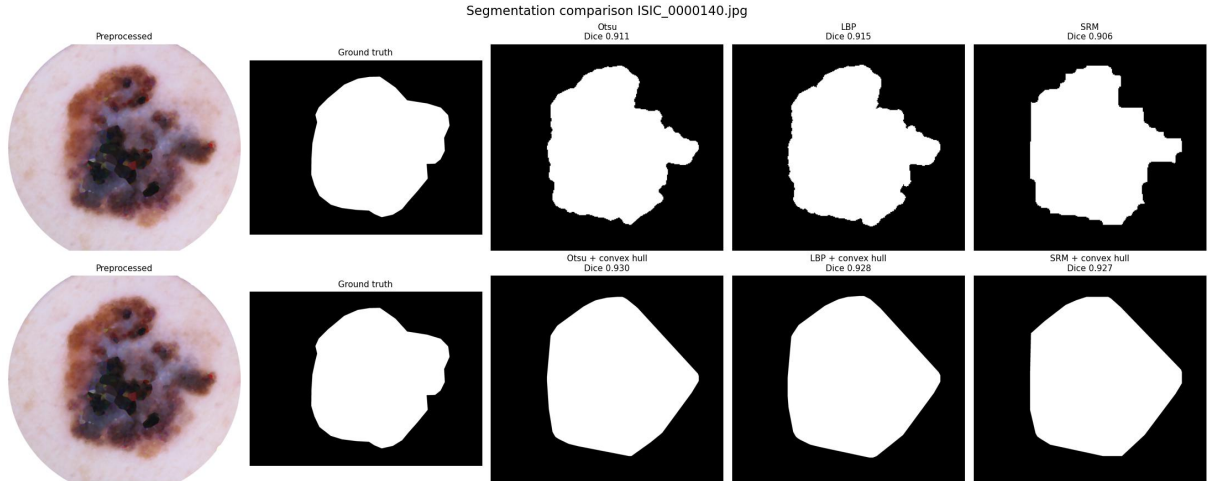


Figure 6: The three methods against the ground truth on one melanoma, with and without the convex hull.

3.4 Limitations

These numbers should be read with their limits in mind.

- **Twenty images is a demonstration, not an evaluation.** No confidence interval computed on this sample would be meaningful.
- **Parameters were tuned on these same images.** There is no held-out set, so the reported scores are optimistic.
- **The inversion guard is a rescue, not a fix.** It flips masks covering more than 60 % of the disc, recovering a failed cluster choice rather than preventing it.
- **A single ground truth per image.** Expert delineations vary between dermatologists, and the Dice ceiling attainable is bounded by that variability, which we cannot measure with one annotation.

Conclusion

We implemented a complete skin-lesion segmentation pipeline using classical computer vision only, from artefact removal through to quantitative evaluation, and compared three methods reimplemented from their source publications.

The outcome contradicts our initial expectation. We had assumed the most sophisticated method, region merging, would dominate; it does not. The texture-based LBP Clustering gives the best raw masks, and the simplest method, multi-channel Otsu, is highly competitive once the convex hull is applied. Algorithmic sophistication did not translate into accuracy on this dataset.

Two findings proved more instructive than the ranking itself. First, the interaction between frame removal and the LBP clustering, where a preprocessing step that fabricates pixel values silently destroyed a method that fits a global colour model, a class of bug invisible in the mean and obvious in the pipeline figures. Second, the cost of smoothing a colour image with a scalar Gaussian, which blurs across the channel axis and quietly corrupted the region scoring. Both were found by inspecting intermediate stages, which is exactly the affordance that a classical pipeline offers and a network does not.

The natural continuations are shading correction [2] to handle heterogeneous illumination, evaluation on a larger ISIC subset with a held-out parameter set, and using these masks as the input to the feature-extraction stage of a full CAD system.

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